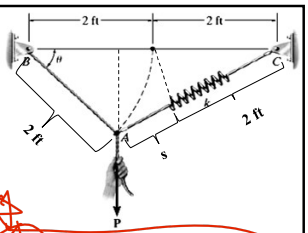
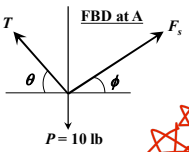
■3–22. A vertical force $P = 10$ lb is applied to the ends of the 2-ft cord AB and spring AC. If the spring has an unstretched length of 2 ft, determine the angle θ for equilibrium. Take $k = 15$ lb/ft.



3-22. 수직 힘 P=10 lb가 2 ft 길이의 코드 AB와 스프링 AC의 끝에 작용한다. 스프링의 변형되지 않은 길이는 2 ft이다. 평형을 만족하는 각도를 구하시오. 스프링 상수는 k = 15 lb/ft로 한다.



Solution

$\sum F_x = 0\ ;\ F_s cos\phi - T cos\theta = 0 \quad \ldots (1)$

$\sum F_y = 0\ ;\ F_s sin\phi + T sin\theta - 10 = 0 \quad \ldots (2)$

$s + 2 = \sqrt{(2sin\theta)^2 + (4 - 2cos\theta)^2}$

$\rightarrow s = \sqrt{4^2 + 2^2 - 2.4.2cos\theta} - 2 = 2\sqrt{5 - 4cos\theta} - 2$

$So, F_s = ks = 2k\left(\sqrt{5 - 4cos\theta} - 1\right)$

From (1), $T = F_s\left(\frac{cos\phi}{cos\theta}\right) = 2k\left(\sqrt{5 - 4cos\theta} - 1\right)\left[\frac{(2-cos\theta)}{\sqrt{5-4cos\theta}}\right]\left[\frac{1}{cos\theta}\right]$

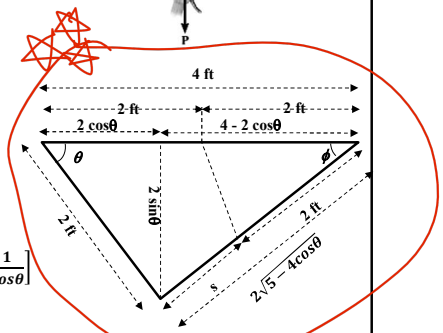
From (2), $2k\left(\sqrt{5 - 4cos\theta} - 1\right)\frac{sin\theta}{\sqrt{5-4cos\theta}}$

$+2k\left(\sqrt{5 - 4cos\theta} - 1\right)\left[\frac{(2-cos\theta)}{\sqrt{5-4cos\theta}}\right]\left[\frac{1}{cos\theta}\right] sin\theta = 10$

$\rightarrow \frac{2k\left(\sqrt{5-4cos\theta}-1\right)}{\sqrt{5-4cos\theta}}(sin\theta + 2tan\theta - sin\theta) = 10$

$\rightarrow \frac{\left(\sqrt{5-4cos\theta}-1\right)}{\sqrt{5-4cos\theta}}(tan\theta) = \frac{10}{4k}$

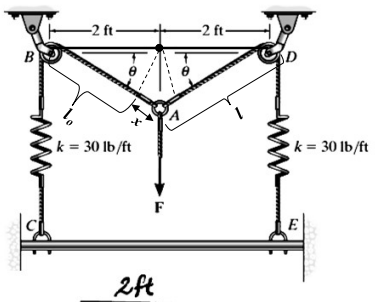
Setting k = 15 lb/ft, we get θ from trial and error method as, $\theta = 35^o$. (Ans.)



$cos\phi = \frac{2(2-cos\theta)}{2\sqrt{5-4cos\theta}} = \frac{(2-cos\theta)}{\sqrt{5-4cos\theta}}$

$sin\phi = \frac{2\,sin\theta}{2\sqrt{5-4cos\theta}} = \frac{sin\theta}{\sqrt{5-4cos\theta}}$

3–30. The springs on the rope assembly are originally unstretched when $\theta = 0°$. Determine the tension in each rope when $F = 90$ lb. Neglect the size of the pulleys at B and D.



3-30. 로프 장치에 있는 스프링들은 θ =0°일 때 원래 길이이다. 힘 F =90 lb일 때 각 로프에 발생하는 장력을 구하시오. B와 D의 도르래 크기는 무시한다.

Solution

$l = \frac{2}{\cos\theta}$

$T = ks = k(l - l_0) = 30\left(\frac{2}{\cos\theta} - 2\right) = 60\left(\frac{1}{\cos\theta} - 1\right) \quad (1)$

$+\uparrow \Sigma F_y = 0; \quad 2T\sin\theta - 90 = 0 \quad (2)$

Substituting Eq. (1) into (2) yields :

$120(\tan\theta - \sin\theta) - 90 = 0$

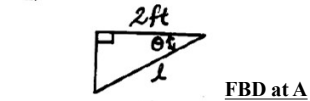
$\tan\theta - \sin\theta = 0.75$

By trial and error :

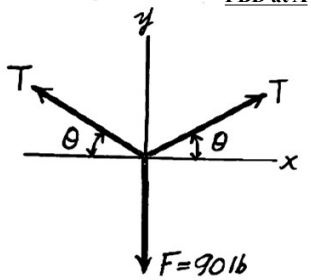
$\theta = 57.957°$

From Eq. (1).

$T = 60\left(\frac{1}{\cos 57.957°} - 1\right) = 53.1$ lb Ans

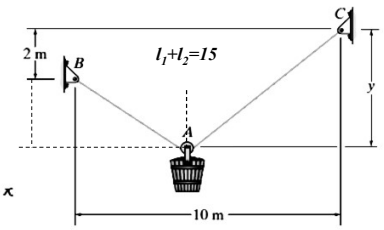


FBD at A



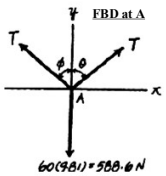
• 3–43. The pail and its contents have a mass of 60 kg. If the cable BAC is 15 m long, determine the distance y to the pulley at A for equilibrium. Neglect the size of the pulley.

3-43. 양동이와 그 내용물의 질량은 60 kg이다.
케이블 BAC의 길이가 15 m일 때, 평형을 만족하도록
도르래 A까지의 거리 y를 구하시오. 도르래의 크기는 무시한다.



Solution

Free Body Diagram : Since the pulley is smooth, the tension in the cable is the same throughout the cable.

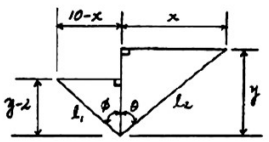


Equations of Equilibrium :

$$\overset{+}{\rightarrow}\Sigma F_x = 0; \quad T\sin\theta - T\sin\phi = 0 \qquad \theta = \phi$$

Geometry :

$$l_1 = \sqrt{(10-x)^2+(y-2)^2} \qquad l_2 = \sqrt{x^2+y^2}$$



Since $\theta = \phi$, two triangles are similar.

$$\frac{10-x}{x} = \frac{y-2}{y} = \frac{\sqrt{(10-x)^2+(y-2)^2}}{\sqrt{x^2+y^2}} \qquad [1]$$

Also,

$$l_1 + l_2 = 15$$

$$\sqrt{(10-x)^2+(y-2)^2}+\sqrt{x^2+y^2} = 15$$

$$\left(\frac{\sqrt{x^2+y^2}}{\sqrt{x^2+y^2}}\right)\sqrt{(10-x)^2+(y-2)^2}+\sqrt{x^2+y^2} = 15 \qquad [2]$$

However, from Eq. [1] $\dfrac{\sqrt{(10-x)^2+(y-2)^2}}{\sqrt{x^2+y^2}} = \dfrac{10-x}{x}$, Eq. [2] becomes

$$\sqrt{x^2+y^2}\left(\frac{10-x}{x}\right)+\sqrt{x^2+y^2} = 15 \qquad [3]$$

Dividing both sides of Eq. [3] by $\sqrt{x^2+y^2}$ yields

$$\frac{10}{x} = \frac{15}{\sqrt{x^2+y^2}} \qquad x = \sqrt{0.8}y \qquad [4]$$

From Eq. [1] $\dfrac{10-x}{x} = \dfrac{y-2}{y}$ $\qquad x = \dfrac{5y}{y-1}$ $\qquad [5]$

Equating Eq. [4] and [5] yields

$$\sqrt{0.8}y = \frac{5y}{y-1}$$

$$y = 6.59 \text{ m} \qquad \text{Ans}$$